

Secure Firmware Update

Secure Firmware Update is supported only on ConnectX-4 onwards adapter cards.

A “Secure firmware update” is the ability of a device to verify digital signatures of new firmware binaries, in order to assure that only officially approved versions can be installed from the host, the network[1] or a Board Management Controller (BMC).

The firmware of devices with “secure firmware up date” functionality (secure FW), restricts access to specific commands and registers that can be used to modify the firmware binary image on the flash, as well as commands that can jeopardize security in general. Most notably, the commands and registers for random flash access are disabled.

Secure FW verifies new binaries before activating them, compared to legacy devices where this task is done by the update tool using direct flash access commands. In addition to signature verification, secure FW also checks that the binary is designated to the same device model, that the new firmware is also secured, and that the new FW version is not included in a forbidden versions blacklist. The firmware rejects binaries that do not match the verification criteria.

Secure FW utilizes the same ‘fail safe’ upgrade procedures, so events like power failure during update should not leave the device in an unstable state. The table below lists the impact of secure FW update on MFT tools.



Tool	Flow	Secure FW	With Token
flint / mlxburn	Burn FW	Working with controlled fw update	Works with controlled fw update
	Query	Working with controlled fw update	Works with controlled fw update
	Set GUIDs	Working with controlled fw update	Works with controlled fw update

		Verify	Working partially (BOOT image)	World part (BOC image)
		Set DV INFO: SET MFG, SET VSD, VPD	Not supported in Secure FW	Not supported in Secure FW
		ROM OPS: BROM, DROM	Not supported, BOOT image modification is not supported (MFBA)	Not supported, BOOT image modification is not supported (MFE)
		"-ocr" override cache replacement (Direct flash GW access)	Not supported in Secure FW	Not supported in Secure FW
		HW SET (Set flash parameters)	Flash GW is blocked	Flash GW is blocked
		"--no_fw_ctrl" (Legacy Flow)	Not supported in Secure FW	Not supported in Secure FW
	mlxfwmanager / mlxup	Burn FW	Working with controlled fw update	Working with controlled fw update
	mlxfwmanager	with --no_fw_ctrl	Not supported in Secure FW	Not supported in Secure FW
	mlxdump	fsdump	Blocked icmds	World part

	phyUc	Blocked icmds	work
	rxdump	CR-Space is locked & Blocked icmds	work
	sxdump	CR-Space is locked & Blocked icmds	work
wqdump	Dump QP contexts	Blocked icmds	work
	Dump WQs	Blocked icmds	work
	ICM	Blocked icmds	work
	WRITE QP (Devmon)		work
mget_temp	hw_access	Read Only CR- Space	work
mcra	Read	working	work
	Write	Read Only CR- Space	work
mstdump	Read	working	work
mlxtrace / fwtrace	MEM & FIFO	Only fwtrace is supported and only in Linux	work

pckt_drop	uses write to CR-Space to work	Read Only CR- Space	work
mlxlink	working	working	work
mlxreg	working	working	work
mlxcables	working	working	work
mlxconfig	working	working	work
mlxfwreset	working	working	work
i2c/mlxi2c	Not relevant when not in livefish		
	With Force flag (ENV VAR)	Read Only CR- Space	work

Secure Firmware Implications on Burning Tools

When Secure Firmware is enabled, the flint output slightly changes due to the differences in the underlying NIC accessing methods. Some functionalities may be restricted according to the device security level.

flint query under secure mode:

```

1  # flint -d /dev/mst/mt4115_pciconf0 q
2  Image type:      FS3
3  FW Version:      12.19.2278
4  FW Release Date: 7.6.2017
5  Description:     UID                               GuidNumber
6  Base GUID:       7cfe90030029205e  4
7  Base MAC:        00007cfe9029205e  4
8  Image VSD:
9  Device VSD:
10  PSID:            MT_2190110032
11  Security Attributes: secure-fw, dev

```

Unavailable information is reported as N/A.



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